

Assessing Regional Climate Disparities under Solar Geoengineering

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Abstract

The escalating impacts of anthropogenic climate change, characterized by a 1.2°C rise in global mean surface temperatures above pre-industrial levels and projections of exceeding 4°C by 2100 under high-emission scenarios, underscore the urgent need for innovative mitigation strategies. Solar Radiation Management (SRM), a form of solar geoengineering, offers a potential intervention by reflecting incoming solar radiation to counteract greenhouse gas-induced warming. Utilizing outputs from the Geoengineering Model Intercomparison Project (GeoMIP) Phases 5 and 6, this study assesses the regional climate disparities induced by SRM, with a particular focus on temperature, precipitation, and atmospheric circulation changes under the G1 and G6 scenarios, compared against 4xCO₂ and RCP8.5 conditions. Results indicate that SRM effectively reduces global warming to a residual range of 0.5–4°C, with pronounced cooling in Asia and Africa, yet persistent warming in Northern Hemisphere high latitudes and the Arctic due to unmitigated polar feedbacks. Precipitation responses reveal a weakened hydrological cycle, with significant reductions (e.g., >1.5 mm day⁻¹ in northern India under SAI) contrasting with increases under high-CO₂ scenarios, driven by altered meridional circulation and subsidence effects. The Indian Summer Monsoon emerges as a critical vulnerability, with SRM-induced drought risks threatening regional water security. These disparities highlight SRM's trade-offs, necessitating region-specific governance to mitigate inequities. This term paper synthesizes multi-model insights to inform policy, advocating for higher-resolution modeling and socio-economic integration to refine SRM's role in a balanced climate strategy.

1. Introduction

The intensifying impacts of anthropogenic climate change—manifested through rising global temperatures, escalating extreme weather events, and widespread hydrological disruptions—pose existential threats to ecosystems, economies, and human societies worldwide. The Sixth Assessment Report of the Intergovernmental Panel on Climate Change (IPCC AR6) underscores that global mean surface temperatures have already increased by approximately 1.2°C above pre-industrial levels, with projections indicating a potential exceedance of 4°C by 2100 under high-emission scenarios like RCP8.5 (IPCC, 2021). Despite international commitments under the Paris Agreement to limit warming to well below 2°C, preferably 1.5°C, greenhouse gas (GHG) emissions continue to rise, driven by persistent reliance on fossil fuels and inadequate global mitigation efforts (Rogelj et al., 2016; Vioni et al., 2023). This trajectory risks triggering irreversible tipping points, such as the collapse of the Greenland and Antarctic ice sheets, permafrost thaw, and disruption of critical monsoon systems, with cascading consequences for global climate stability (Lenton et al., 2008; Irvine et al., 2009). Against this backdrop, solar geoengineering—specifically Solar Radiation Management (SRM)—has emerged as a contentious yet increasingly studied intervention to temporarily stabilize global temperatures while decarbonization efforts are scaled up (Caldeira et al., 2013; Vioni et al., 2023).

SRM encompasses a suite of techniques designed to counteract GHG-induced warming by reflecting a small fraction of incoming solar radiation back to space, thereby increasing Earth's albedo and reducing the net radiative forcing at the surface (Caldeira et al., 2013). Unlike carbon dioxide removal (CDR), which addresses the root cause of warming by extracting CO₂ from the atmosphere, SRM acts as a symptomatic treatment, offering rapid cooling potential but leaving atmospheric CO₂ concentrations unaltered (Matthews & Caldeira, 2007). Proposed SRM methods include stratospheric aerosol injection (SAI), which involves dispersing reflective particles—typically sulfur-based—into the stratosphere to mimic the cooling effects of volcanic eruptions like Mount Pinatubo in 1991 (Crutzen, 2006; Robock et al., 2008); marine cloud brightening (MCB), which enhances the reflectivity of low-level clouds by increasing cloud condensation nuclei (Latham et al., 2008); surface albedo modification, such as whitening urban areas or crops (Akbari et al., 2009; Ridgwell et al., 2009); and space-based reflectors, though these remain technologically speculative (Angel, 2006). SAI, in particular, has been the focus of extensive

modeling due to its theoretical scalability and relatively low deployment costs compared to CDR alternatives (Rasch et al., 2008b).

Modeling studies suggest that SRM, especially SAI, could reduce global mean temperatures to near pre-industrial levels within months to years, offering a potential buffer against acute climate impacts (Matthews & Caldeira, 2007; Wigley, 2006). For instance, Geoengineering Model Intercomparison Project (GeoMIP) simulations, such as the G1 experiment (solar dimming to offset a quadrupling of CO₂), demonstrate SRM's capacity to neutralize global warming, albeit with residual regional anomalies (Kravitz et al., 2011; Tilmes et al., 2013). However, this global efficacy obscures significant regional disparities, particularly in hydrological cycles, atmospheric circulation, and socio-economic outcomes, which form the crux of this paper's inquiry.

While SRM's ability to lower global temperatures is well-documented, its deployment introduces novel climate states that differ markedly from both pre-industrial and high-CO₂ scenarios due to fundamental differences in radiative forcing mechanisms. GHGs primarily trap outgoing longwave radiation, warming the troposphere and surface, whereas SRM reduces incoming shortwave radiation, cooling the surface but altering stratospheric dynamics and moisture fluxes (Schmidt et al., 2012; Modak et al., 2016). This asymmetry results in a climate that cannot fully replicate pre-industrial conditions, even if global mean temperatures are balanced (Caldeira et al., 2013). A paramount concern is SRM's impact on the global hydrological cycle, which is intricately linked to evaporation, precipitation, and atmospheric circulation patterns.

Research from Geoengineering Model Intercomparison Project (GeoMIP) and other modeling efforts consistently indicates that SRM suppresses the hydrological cycle, reducing global mean precipitation and evaporation by approximately 4.5% compared to pre-industrial levels (Tilmes et al., 2013; Irvine et al., 2016). This suppression arises from decreased solar input, which reduces surface evaporation, and CO₂-physiological effects, where elevated CO₂ levels cause plants to close stomata, further limiting evapotranspiration (Bala et al., 2008; Cao et al., 2010). However, these global averages mask pronounced regional variations. Tropical monsoon regions, such as South Asia, West Africa, and South America, face significant precipitation reductions—up to 6–7% in some GeoMIP scenarios—due to weakened convective activity and altered moisture transport (Tilmes et al., 2013; Bhowmick et al., 2021). Conversely, mid-latitude

regions may experience increased precipitation, leading to a “wet-get-drier, dry-get-wetter” pattern driven by shifts in atmospheric circulation, such as changes in the Hadley Cell and Intertropical Convergence Zone (ITCZ) (Simpson et al., 2019; Vioni et al., 2023).

These hydrological disparities are compounded by SRM’s influence on atmospheric circulation. SAI, for example, induces stratospheric heating, which stabilizes the troposphere, suppresses convection, and alters large-scale circulation patterns like the Tibetan Anticyclone and jet streams (Ferraro et al., 2014; Simpson et al., 2019). Vioni et al. (2023) highlight that even targeted SAI deployments—such as injections confined to specific hemispheres—can disrupt inter-hemispheric energy balances, shifting the ITCZ and exacerbating precipitation asymmetries. Such shifts could intensify drought in already water-stressed regions like the Sahel while increasing flood risks in others, such as parts of Southeast Asia (Jones et al., 2018). Moreover, SRM does not mitigate CO₂-driven effects like ocean acidification or changes in terrestrial carbon cycles, further complicating regional climate responses (Matthews et al., 2009).

The regional disparities induced by SRM carry profound socio-economic and ethical implications, particularly for vulnerable populations in developing nations. South Asia, home to over 1.4 billion people, relies heavily on the Indian Summer Monsoon (ISM) for more than 70% of its annual precipitation, supporting agriculture, water resources, and economic stability (Bhowmick et al., 2021). GeoMIP studies suggest that SAI could weaken the ISM, reducing rainfall by up to 6% and increasing the risk of monsoon failure, with severe consequences for food security and livelihoods (Tilmes et al., 2013; Bhowmick et al., 2021). Similar risks are projected for West African and South American monsoons, where precipitation reductions could disrupt ecosystems and exacerbate poverty (Jones et al., 2018; Trisos et al., 2018). These outcomes highlight the potential for SRM to exacerbate global inequities, as regions least responsible for historical emissions may bear disproportionate burdens (Rahman et al., 2018).

Ethically, SRM raises concerns about governance, consent, and the risk of unilateral deployment. The “termination shock”—a rapid warming event triggered by abrupt cessation of SRM—poses a particularly acute risk, as it could release stored carbon from oceans and land, overwhelming adaptation capacities in vulnerable regions (Matthews & Caldeira, 2007; Robock et al., 2008). Furthermore, SRM’s reliance on continuous intervention creates an intergenerational

dependency, transferring risks to future generations if maintenance falters (Caldeira et al., 2013). These challenges underscore the need for rigorous scientific evaluation to inform equitable and transparent governance frameworks.

The motivation for this study lies in the urgent need to move beyond global metrics and critically examine SRM's regional impacts, particularly in climate-sensitive areas like South Asia. While SRM's potential to avert catastrophic warming is compelling, its propensity to create regional disparities—especially in precipitation and circulation—demands thorough scrutiny to avoid unintended harm. This term paper focuses on assessing these disparities using GeoMIP5 and GeoMIP6 outputs, with a particular emphasis on the hydrological cycle, atmospheric circulation changes, and their implications for the Indian Summer Monsoon and South Asia's broader climate system.

South Asia's selection as a focal region is driven by its demographic, economic, and climatic significance. The ISM's sensitivity to radiative forcing and circulation changes makes it a critical test case for evaluating SRM's regional effects (Bhowmick et al., 2021). Minor deviations in monsoon rainfall can disrupt agriculture, water security, and economic stability, affecting millions of lives (Fasullo, 2012). By analyzing temperature, precipitation, and circulation anomalies—specifically wind fields at 200 hPa and 850 hPa—this study aims to elucidate the mechanisms driving SRM-induced disparities and their socio-economic ramifications.

The paper's contribution lies in its integrative approach, synthesizing GeoMIP findings to provide a nuanced understanding of SRM's regional impacts. It builds on seminal works like Tilmes et al. (2013), which highlighted global hydrological suppression, and Vioni et al. (2023), which underscored circulation-driven asymmetries. By focusing on South Asia, the study addresses a gap in region-specific analyses, offering insights into SRM's feasibility and risks in a high-stakes context. Furthermore, it engages with equity and governance considerations, emphasizing the need for inclusive frameworks to navigate SRM's complex trade-offs.

Research Objectives

This term paper pursues the following objectives to rigorously assess SRM's regional impacts:

1. **Evaluate Temperature and Precipitation Responses:** Quantify global and regional changes in temperature and precipitation under SRM scenarios using GeoMIP5 (e.g., G1) and GeoMIP6 (e.g., G6solar, G6sulfur) outputs, focusing on disparities between tropical and mid-latitude regions.
2. **Analyze Circulation Changes:** Investigate alterations in atmospheric circulation patterns that drive precipitation shifts in mid-latitude and tropical zones under geoengineered climates.
3. **Identify Regional Benefits and Risks:** Highlight regions experiencing significant benefits (e.g., reduced extreme heat) versus risks (e.g., precipitation deficits or residual warming), with a focus on vulnerable areas like South Asia, Africa, and South America.
4. **Assess Impacts on the Indian Summer Monsoon:** Specifically evaluate the effects of SAI analogues on the ISM and South Asia's climate, analyzing precipitation, temperature, and circulation anomalies to understand implications for water security and agriculture.
5. **Inform Policy and Equity Considerations:** Discuss the socio-economic and ethical implications of SRM-induced disparities, proposing governance principles to ensure equitable outcomes and mitigate risks like termination shock.

To achieve these objectives, the paper is organized as follows: Section 2 reviews the literature on SRM's hydrological impacts, circulation changes, and regional disparities, grounding the analysis in GeoMIP and related studies. Section 3 details the methodology, including GeoMIP datasets, variables (temperature, precipitation, wind fields), and analytical approaches (e.g., anomaly mapping, correlation analysis). Section 4 presents results, detailing global and regional climate responses, with a dedicated focus on South Asia and the ISM. Section 5 interprets these findings, comparing them to existing literature, exploring circulation-precipitation linkages, and addressing policy and equity implications. Section 6 concludes by summarizing key insights, highlighting governance challenges, and suggesting avenues for future research to refine SRM's role in climate strategy. This comprehensive approach ensures a thorough examination of SRM's regional impacts, contributing to the scientific and policy discourse on managing climate risks in an increasingly uncertain world.

2. Literature Review

This review synthesizes recent research on the hydrological impacts of solar radiation management (SRM), a geoengineering strategy proposed to mitigate global warming by reducing incoming solar radiation. Drawing primarily from studies within the GeoMIP, this analysis evaluates the potential benefits and risks of SRM, focusing on its effects on global and regional precipitation, evaporation, and atmospheric circulation. The reviewed literature, spanning 2011 to 2025, addresses the scientific, technical, and policy implications of SRM, highlighting its limitations and uncertainties.

Motivation and Background

The urgency to address climate change, driven by persistent rises in greenhouse gas (GHG) emissions despite international mitigation efforts like the Paris Agreement, has fueled interest in SRM. Techniques such as stratospheric sulfate aerosol injection and solar irradiance reduction aim to counteract warming by mimicking the cooling effects observed after volcanic eruptions (e.g., Mount Pinatubo). However, while SRM can lower global temperatures, prior studies suggest it may disrupt the hydrological cycle, altering precipitation patterns and posing regional risks (Tilmes et al., 2013; Irvine et al., 2016). GeoMIP provides a standardized framework for multi-model comparisons to assess these impacts, addressing inconsistencies in earlier research and facilitating collaboration among modeling groups and policymakers (Kravitz et al., 2011; Visionsi et al., 2023). The background of these studies underscores a critical tension: SRM offers a potential rapid response to warming but introduces novel climate states with uncertain side effects. High-emission scenarios (e.g., RCP8.5) and the failure of traditional mitigation strategies have intensified the need to evaluate SRM's efficacy and risks, particularly for vulnerable regions reliant on monsoon systems and seasonal rainfall (Schmidt et al., 2012; Bhowmick et al., 2021).

Objectives and Methodological Approaches

The reviewed studies share objectives to assess SRM's hydrological impacts, compare modeling outcomes, and identify uncertainties. Key goals include:

1. Evaluating changes in global and regional precipitation, evaporation, and moisture fluxes under SRM scenarios compared to high-CO₂ and pre-industrial climates (Tilmes et al., 2013; Schmidt et al., 2012).
2. Investigating SRM's ability to restore hydrological cycles to pre-industrial levels and its regional disparities, especially in tropical monsoon regions (Irvine et al., 2016; Bhowmick et al., 2021).
3. Quantifying the role of stratospheric heating and aerosol-specific processes (e.g., radiative forcing, circulation changes) in driving hydrological responses (Ferraro et al., 2014; Simpson et al., 2019).
4. Standardizing experimental designs to enable robust intermodel comparisons and inform policy (Kravitz et al., 2011; Vioni et al., 2023).

Methodologically, these studies employ global climate models (GCMs) within the GeoMIP framework, simulating scenarios mentioned in Figure 1. Some also compare SRM to idealized sunshade geoengineering, highlighting differences in atmospheric dynamics and chemical feedbacks (Modak et al., 2016; Jones et al., 2018).

Novel Findings

Globally, SRM reduces mean precipitation and weakens the hydrological cycle, primarily due to decreased surface evaporation from lower solar radiation (Tilmes et al., 2013; Irvine et al., 2016). This suppression is more pronounced with stratospheric aerosol injection than with solar dimming, as aerosols induce stratospheric heating that stabilizes the troposphere, suppresses convection, and alters circulation patterns (Simpson et al., 2019).

Regionally, tropical monsoon areas, such as South Asia and Africa, experience significant drying, raising concerns about water availability and socio-economic stability (Bhowmick et al., 2021; Jones et al., 2018). For instance, the Indian Summer Monsoon strengthens under global warming due to enhanced land-ocean temperature gradients, but SRM reverses this effect, reducing precipitation and increasing monsoon failure risks (Bhowmick et al., 2021). Conversely, some midlatitude regions may see increased precipitation, creating a “wet-get-drier, dry-get-wetter” pattern driven by circulation shifts (Simpson et al., 2019).

3. Materials and Methods

The Geoengineering Model Intercomparison Project (GeoMIP) Framework

To investigate the regional climate disparities induced by solar geoengineering, this study leverages the standardized modeling framework provided by the GeoMIP, initiated in 2010 as an extension of the Coupled Model Intercomparison Project (CMIP) (Kravitz et al., 2011). GeoMIP facilitates robust, multi-model evaluations of SRM scenarios by coordinating experiments across international climate modeling centers, thereby minimizing inconsistencies inherent in single-model studies and enhancing the reliability of projections (Visioni et al., 2023). The GeoMIP framework includes a suite of standardized simulations designed to explore the physical processes and climatic impacts of SRM, with a focus on techniques such as SAI and solar dimming (Kravitz et al., 2015). These experiments, conducted using fully coupled Atmosphere-Ocean General Circulation Models (AOGCMs), span a 50-year simulation period to capture both transient and equilibrium climate responses (Kravitz et al., 2011).

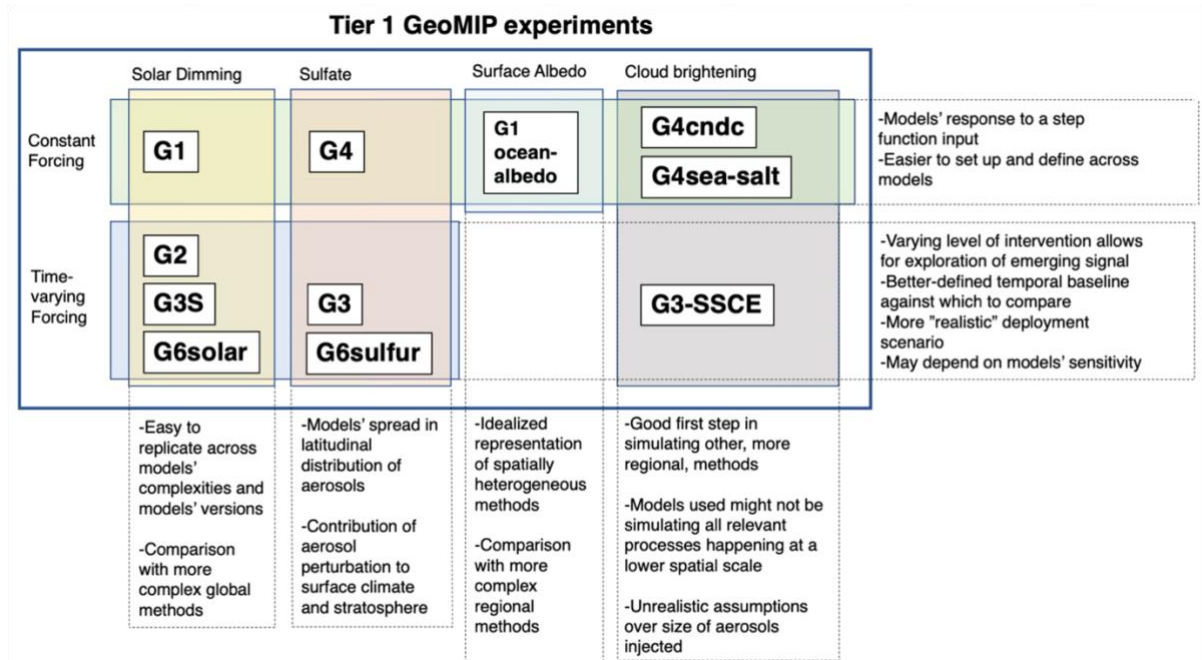


Figure 1: Schematic summary of all Tier-1 GeoMIP experiments across different iterations

GeoMIP's experimental protocols are meticulously designed to isolate SRM's effects on key climate variables, including temperature, precipitation, and atmospheric circulation, under controlled scenarios of radiative forcing. The present analysis utilizes outputs from GeoMIP5,

specifically the G1 experiment, complemented by reference simulations (piControl and 4xCO₂), as these provide a clear framework for comparing geoengineered climates against pre-industrial and high-CO₂ baselines (Tilmes et al., 2013). By employing an ensemble approach, GeoMIP mitigates uncertainties arising from model-specific parameterizations, such as convection schemes, aerosol microphysics, and cloud feedbacks, ensuring more robust and generalizable findings (Schmidt et al., 2012; Vioni et al., 2021b).

G1 Experiment. Of all possible experiments, the simplest and easiest to replicate in different climate models aims to offset the radiative forcing from an increase in CO₂ with a reduction in the model's solar constant. This method directly represents the idea of space sunshades (Angel, 2006): while potentially effective, technical feasibility and costs associated with deployment presently make such an approach prohibitive when compared to other proposed climate geoengineering methods (e.g., The Royal Society, 2009). More relevant for immediately practical geoengineering methods, solar dimming approximates the broad radiative effects of stratospheric aerosol injection. Experiments G1 (CMIP5) and G1ext (CMIP6) involved offsetting the forcing from an instantaneous quadrupling of the CO₂ concentration (abrupt4xCO₂, a standard CMIP experiment) with solar constant reduction, so that the net top-of-atmosphere (TOA) radiative flux in each model was within $\pm 0.1 \text{ W m}^{-2}$ of the baseline value in the first decade of simulation (Kravitz et al., 2011). The high level of replicability means that results between CMIP5 and CMIP6 could be easily compared (Kravitz et al., 2021), even in some cases allowing for a comparison between different model versions. G1 has been extensively studied in terms of the hydrological response (Tilmes et al., 2013; Kravitz et al., 2013, 2014) and from an energetic and thermodynamics perspective (Russotto and Ackerman, 2018a, b; Virgin and Fletcher, 2022), highlighting both some commonalities in the response of the hydrological cycle to a reduction in incoming shortwave radiation and some large discrepancies in the cloud response. Results from G1 have been used for more specific impact analyses (e.g., Bal et al., 2019; Harding et al., 2020), which poses two important issues. First, G1 is an extreme, idealized case, and thus a straightforward analysis of climate model output from G1 cannot be used as a prediction of what climate engineering would do under any practical deployment strategy. This is particularly true for variables such as regional precipitation for which solar dimming is a poor proxy for aerosol injections or other methods (see Niemeier et al., 2013; Vioni et al., 2021a) or for comparisons of tropical and high-latitude effects,

considering that uniformly reducing the solar constant tends to produce a stronger cooling in the tropics (e.g., Govindasamy et al., 2003; Kravitz et al., 2013). Second, there is no single answer for what climate engineering “would do”, as the effects of climate engineering can, to some degree, be designed to mitigate any residual climate change impact

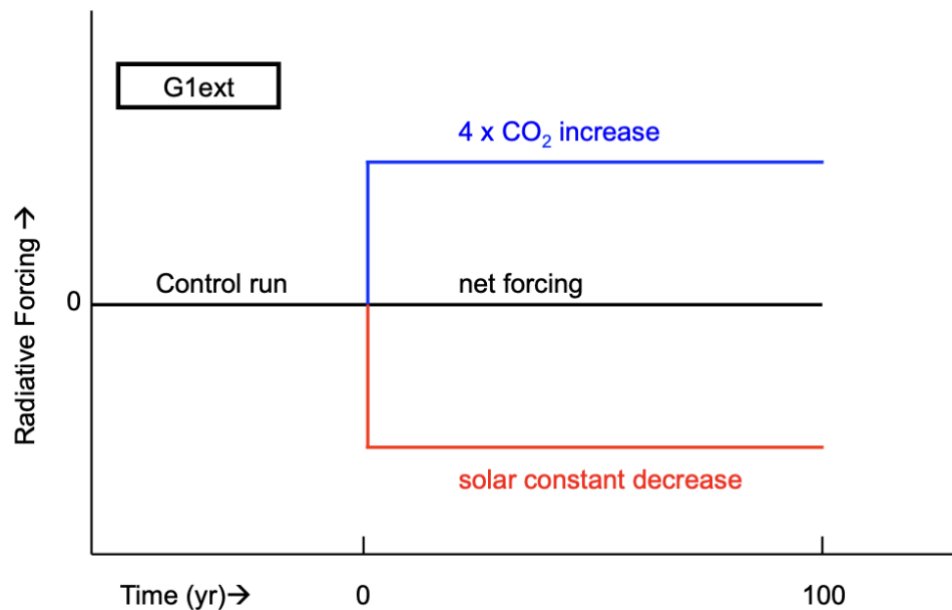


Figure 2: Schematic of experiment G1ext. The experiment is started from a preindustrial control run. The instantaneous quadrupling of the CO₂ concentration from its preindustrial value is balanced by a reduction in solar irradiance for 100 years.

G6Sulfur experiment. Previous GeoMIP experiments (G3 and G4) used RCP4.5 (Representative Concentration Pathway 4.5) as a background scenario. To maintain relevance to the newly designed experiments in CMIP6, our background scenario is changed to follow the ScenarioMIP Tier 1 scenarios, described above. Under experiment G6sulfur (Fig. 2), stratospheric sulfate aerosol precursors will be injected into the model with the goal of reducing the magnitude of the net anthropogenic radiative forcing from the ScenarioMIP Tier 1 high forcing scenario to match that of the ScenarioMIP Tier 1 medium forcing scenario. The motivation for this choice is to evaluate a climate in which geoengineering is used to only partially offset climate change, which would hopefully reduce the burden of adaptation. The choice of the medium forcing scenario as the target, instead of the low forcing scenario, is because the required amount of sulfate aerosol injection to achieve a low anthropogenic forcing is quite large. Representing such large values of injection in a variety of climate models will likely lead to highly variable inter-model results that

are overly sensitive to individual parameterizations. For this experiment, geoengineering will be simulated over years 2020–2100. All atmospheric constituents in the ScenarioMIP Tier 1 scenarios are well defined through the year 2100. Some modeling groups may have an internal sulfate aerosol treatment; the radiative response to stratospheric sulfate aerosol injection should be determined in each model so the proper amount of aerosol is injected into the model such that the total radiative forcing objectives of the experiment are met. This procedure will be more difficult for models that have a complex microphysical treatment of the aerosols, which may require more sophisticated methods of meeting the goals of G6sulfur. One method to calculate the necessary amount of sulfate aerosol is a double radiation call, once with and once without the stratospheric aerosols. Another potential method involves using feedback methods (Jarvis and Leedal, 2012; Kravitz et al., 2014b; MacMartin et al., 2014). The amount of sulfate injection needed for a given model to achieve the goals of this experiment may vary, so modeling groups should scale the aerosol and ozone perturbation fields as necessary.

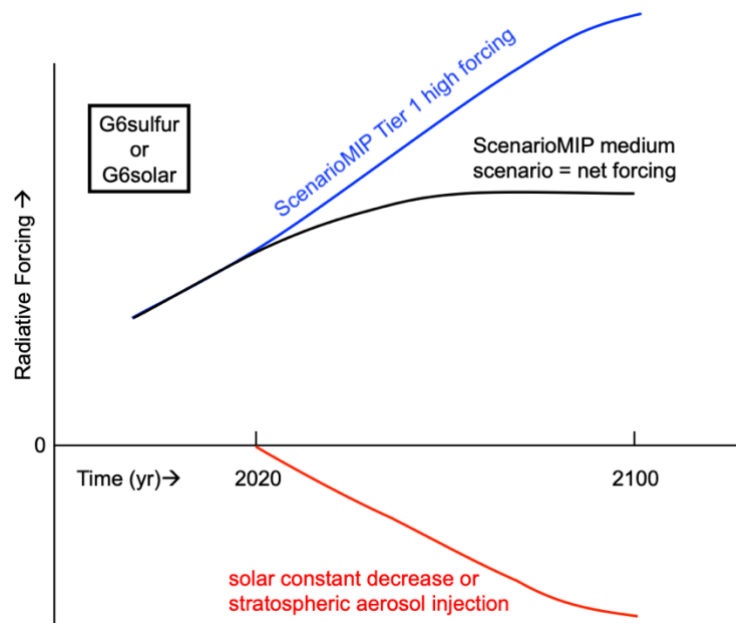


Figure 3: Schematic of experiments G6sulfur and G6solar. Against a background of the ScenarioMIP Tier 1 high forcing scenario, geo- engineering will be conducted at time-varying amounts to return net anthropogenic radiative forcing to the levels of the ScenarioMIP Tier 1 medium forcing scenario. Geoengineering will be accomplished by stratospheric aerosol injection (G6sulfur) or solar irradiance reduction (G6solar).

G6Solar experiment. Experiment G3solar was proposed as an unofficial counter- part to experiment G3 (Kravitz et al., 2011; Table 1); in G3solar, the goals of G3 were achieved using a solar irradiance reduction rather than stratospheric sulfate aerosol injections. Comparison of these

two simulations would reveal differential effects of sulfate aerosols and solar irradiance reduction. Preliminary results from a limited set of models show some differences in the results of the two experiments, particularly related to the hydrological cycle response (Niemeier et al., 2013). We propose G6solar as a parallel experiment to G6sulfur, to compare the effects of solar reduction with those of stratospheric aerosols. G6solar uses the same setup as G6sulfur, but geoengineering is performed using solar irradiance reduction (Fig. 3). In particular, the inter-model differences in the spatial distribution of forcing are likely to be smaller than in G6sulfur, providing a useful context on the effects of uncertainties in stratospheric sulfate aerosol transport.

Description of simulations

We analyze four sets of simulations from 2020 to 2100: two baseline scenarios without geoengineering that follow two Shared Socioeconomic Pathways, SSP2-4.5 and SSP5-8.5 (O'Neill et al., 2016), and two scenarios with geoengineering, G6solar and G6sulfur (Kravitz et al., 2015). Overall, six models participated in all experiments (Table 1).

In the SSP2-4.5 and SSP5-8.5, GHG emissions follow a medium and high trajectory, respectively, resulting by the end of the century in a radiative forcing indicated by the last two numbers in the name (i.e., 4.5 and 8.5 W/m², similar to the Representative Concentration Pathways in CMIP5). The G6 simulations start in 2020 with the same emissions as SSP5-8.5 and, on top of that, have either the solar constant reduced by a certain fraction or produce a sulfate aerosol optical depth with the aim of reducing the globally averaged surface temperature down to the SSP2-4.5 level. Two models (IPSL-CM6A-LR and UKESM1-0-LL) injected SO₂ uniformly between 10° N and 10° S between 18 and 20 km of altitude and across a single longitudinal band (° 0). CESM2(WACCM) injected SO₂ at the Equator and at 25 km of altitude. The others prescribed an already-calculated aerosol optical depth distribution: CNRM-ESM2-1 used an input dataset provided by GeoMIP (the aerosol distribution the G4SSA experiment; Tilmes et al., 2015), while MPI-ESM prescribed their own aerosol distribution derived from the simulations described in Niemeier and Schmidt (2017) and Niemeier et al. (2020). Therefore, the response of direct and diffuse radiation at the surface and the localized stratospheric warming due to radiative heating are fully consistent with those from the other models where the full aerosol production from SO₂ is simulated.

Table 1: Models used and their configurations

Model name and CMIP6 DOI (resolution)	SSP2-4.5	SSP5-8.5	G6solar	G6sulfur	Stratospheric aerosols in G6sulfur	Interactive stratospheric ozone
CESM2(WACCM) Danabasoglu (2019) h:288 × 192, v:70	2 r1, r2	2 r1, r2	2 r1, r2	2 r1, r2	From SO ₂ injection ¹	Yes
CNRM-ESM2-1 Seferian (2018) h:256 × 128, v:40	3 r1, r2, r3	3 r1	1 r1	3 r1, r2, r3	AOD scaled from Tilmes et al. (2015)	Yes
IPSL-CM6A-LR Boucher et al. (2018) h:144 × 143, v:79	1 r1	1 r1	1 r1	1 r1	From SO ₂ injection	No
MPI-ESM1.2-LR Wieners et al. (2019) h:192 × 96, v:47	3 r1, r2, r3	3 r1, r2, r3	3 r1, r2, r3	3 r1, r2, r3	AOD scaled from Niemeier and Schmidt (2017)	No
MPI-ESM1.2-HR Jungclaus et al. (2019) h:384 × 192, v:95	3 r1, r2, r3	3 r1, r2, r3	3 r1, r2, r3	3 r1, r2, r3	AOD scaled from Niemeier and Schmidt (2017)	No
UKESM1-0-LL Tang et al. (2019) h:192 × 144, v:85	3 r1, r4, r8	3 r1, r4, r8	3 r1, r4, r8	3 r1, r4, r8	From SO ₂ injection	Yes

¹ Injected at the Equator at 25 km in deviation from the protocol described by Kravitz et al. (2015).

For the first part of the study, the analysis focuses on three GeoMIP5 simulation subsets, each designed to isolate specific aspects of SRM's climatic impacts:

1. **piControl (CTL):** This serves as the baseline simulation, representing a pre-industrial climate with steady-state CO₂ concentrations (approximately 280 ppm). Models are initialized from a long-term control run that has reached radiative equilibrium, ensuring a

stable reference for anomaly calculations (Kravitz et al., 2011). The CTL simulation provides a benchmark for assessing deviations induced by both high-CO₂ and geoengineered scenarios.

2. **4xCO₂**: This experiment simulates an abrupt quadrupling of atmospheric CO₂ concentrations (to approximately 1120 ppm) from pre-industrial levels, maintaining this level for 50 years. The 4xCO₂ scenario represents a high-emission, unmitigated climate, allowing evaluation of SRM's capacity to offset extreme warming and associated hydrological changes (Schmidt et al., 2012). It induces a radiative forcing increase of approximately 7–8 W/m², serving as the counterfactual against which SRM interventions are measured.
3. **G1**: The G1 experiment tests SRM's efficacy by applying a uniform reduction in solar insolation to offset the top-of-atmosphere (TOA) radiative imbalance caused by the 4xCO₂ scenario. The insolation reduction is calibrated such that the global mean TOA radiative difference between G1 and CTL remains below 0.1 W/m² over the first 10 years of the 50-year simulation (Kravitz et al., 2011). This idealized approach mimics the radiative effects of SAI without the complexities of aerosol microphysics, enabling a clear assessment of SRM's global and regional impacts (Tilmes et al., 2013).

Each experiment is conducted for 50 years and the multi-model ensembles are constructed by averaging outputs across the eight models, reducing noise from internal variability and model-specific biases (Schmidt et al., 2012).

The primary variables analyzed include:

- **Surface Air Temperature**: Monthly mean 2-meter air temperature, used to assess SRM's cooling efficacy and identify residual warming or overcooling across regions.
- **Precipitation**: Monthly total precipitation (convective and large-scale), critical for evaluating hydrological cycle changes, particularly in monsoon-dependent regions like South Asia.

- **Wind Fields:** Zonal and meridional wind components at 200 hPa (upper troposphere, reflecting jet stream and anticyclone dynamics) and 850 hPa (lower troposphere, indicative of monsoon flows and moisture transport). These levels capture circulation changes driving regional precipitation anomalies (Simpson et al., 2019).

Data are regridded to a common latitude-longitude grid to ensure consistency across models, using conservative interpolation to preserve physical quantities (Jones et al., 2018). Anomalies are computed as differences between experimental runs (4xCO₂ – CTL, G1 – CTL) to highlight the impacts of high-CO₂ and geoengineered climates relative to pre-industrial conditions.

Focus on the Indian Summer Monsoon. Given South Asia’s vulnerability to hydrological disruptions, a dedicated analysis targets the ISM’s response to SRM. The ISM is defined over the region 5°–35°N, 60°–100°E, encompassing India, Pakistan, Bangladesh, and parts of the Himalayan foothills (Fasullo, 2012). Seasonal precipitation anomalies are computed for June–September, capturing the monsoon’s core period. Wind anomalies at 850 hPa are examined to assess changes in low-level monsoon flows, while 200 hPa winds are analyzed to evaluate shifts in the Tibetan Anticyclone, a critical driver of monsoon dynamics (Bhowmick et al., 2021).

Data Sources and Availability. All GeoMIP5 datasets are publicly available through the Earth System Grid Federation (ESGF) nodes. Model outputs are stored in NetCDF format, processed using Python libraries (xarray, cartopy), NCAR Command Language (NCL) and the Climate Data Operators (CDO) toolkit for regridding, anomaly calculations, and statistical analysis.

4. Results

Global Surface Air Temperature

The application of solar geoengineering via a uniform reduction in solar insolation in the G1 experiment effectively mitigates a substantial portion of the warming induced by the 4xCO₂ scenario (Figure 4, panels c1 and c2). The multi-model mean temperature anomaly (G1 – CTL) reveals a significant reduction in global mean warming, with residual temperatures ranging from 0.5°C to 4°C above pre-industrial levels. This residual warming is statistically significant at the

98% confidence level (stippled regions) and exhibits pronounced spatial heterogeneity, with greater persistence over continental landmasses compared to oceanic regions, reflecting differential heat storage capacities (Schmidt et al., 2012).

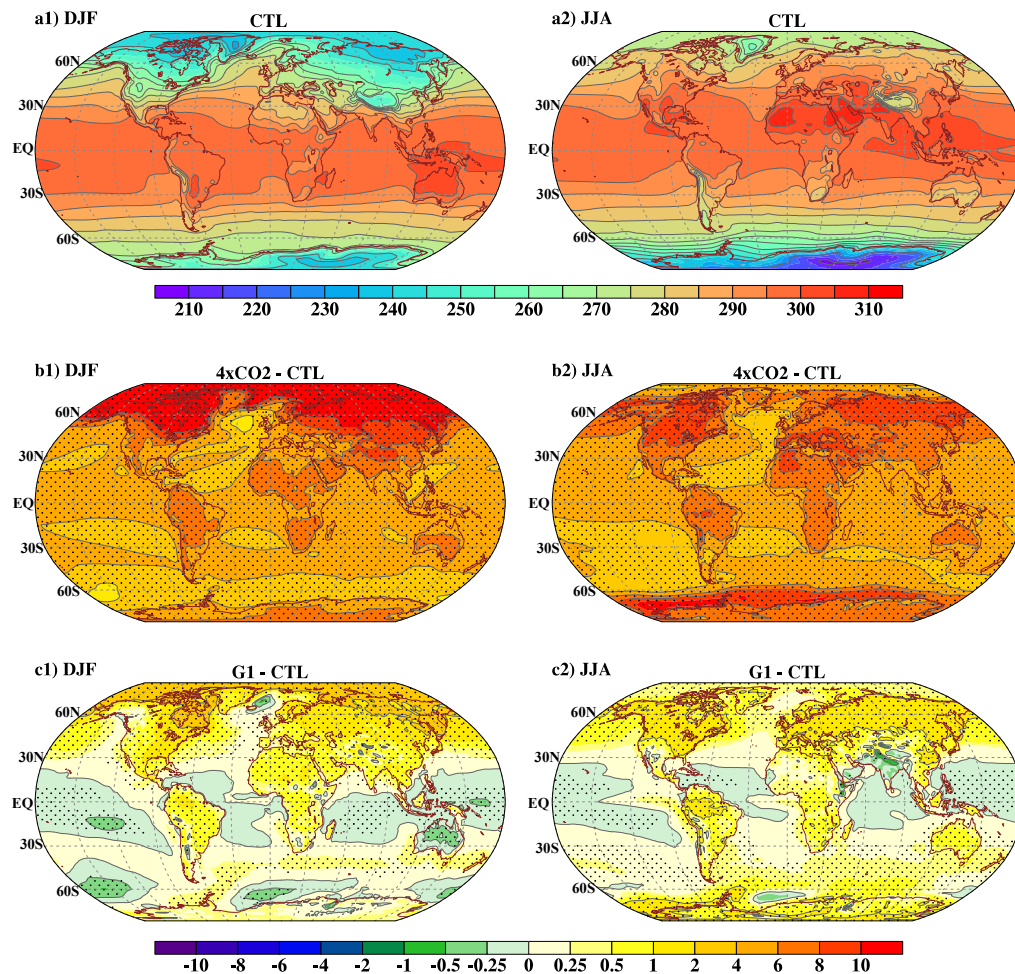


Figure 4: Spatial distribution of changes in surface air temperature in 4xCO₂ and G1 simulations w.r.t. control climate. (a1,a2) Mean surface temperature in control climate during (a1) winter and (b1) summer. (b1,b2) Differences in surface temperature between 4CO₂ and control during (b1) winter and (b2) summer from multi model mean of eight models of GeoMIP5. (c1,c2) Same as (b1,b2) but for G1. Stippling marks the regions of statistically significant differences (see the methodology section for the statistical measures).

Notably, NH high latitudes—including Russia, the United States, Canada, and the Arctic—experience persistent residual warming exceeding 1°C across both boreal winter (December–February, DJF) and summer (June–August, JJA) seasons, with the Arctic showing the most pronounced anomalies (>2°C) during DJF (Figure 4, c1). This residual warming is attributed to

incomplete compensation of polar amplification effects, where reduced sea ice and snow cover sustain positive albedo feedbacks despite SRM (Serreze & Barry, 2011). In contrast, Europe, Africa, and South America exhibit marginal residual warming ($<1^{\circ}\text{C}$) across seasons, suggesting a more effective cooling response in these regions, though statistical significance varies spatially.

Regional analysis highlights several key disparities in the geoengineered climate. Over Antarctica, residual warming is observed during JJA (Figure 4, c2), likely due to diminished sea-ice volume in the G1 scenario compared to CTL, a finding consistent with previous studies on SRM-induced polar climate dynamics (Tilmes et al., 2013). In the Indian subcontinent, a complex pattern emerges: significant overcooling (up to -2°C) is evident over the Indo-Gangetic Plain during JJA (Figure 1, c2), despite widespread residual warming ($1\text{--}2^{\circ}\text{C}$) across Asia. This overcooling, statistically significant at the 98% confidence level, may result from enhanced monsoon-related cloud cover and latent heat fluxes, partially offsetting SRM's cooling effect (Bhowmick et al., 2021). Conversely, during DJF (Figure 4, c1), the Indian subcontinent experiences residual warming, indicating seasonal variability in SRM's efficacy.

Oceanic regions exhibit distinct responses, with subcooling observed over tropical oceans in both seasons and southern oceans during DJF. The Northern Indian Ocean shows non-significant cooling during JJA, while the Atlantic exhibits non-significant cooling in both seasons, suggesting that SRM's radiative forcing reduction is less effective in modulating oceanic heat budgets compared to land areas (Visoni et al., 2023). This tropical cooling is consistent with the more efficient reduction of net radiative forcing in low-latitude regions, where insolation is highest, though the lack of statistical significance indicates uncertainty in ocean-atmosphere coupling across models (Kravitz et al., 2013).

The persistent residual warming in NH high latitudes and the Arctic underscores SRM's inability to fully counteract CO_2 -driven feedbacks, such as those associated with ice-albedo and permafrost carbon release (Matthews & Caldeira, 2007). The overcooling in the Indo-Gangetic Plain during JJA suggests potential disruptions to the Indian Summer Monsoon, a critical hydrological system for South Asia, which may alter agricultural productivity and water availability (Bhowmick et al., 2021). The marginal warming in Africa and South America indicates a relatively balanced response, but localized anomalies could still impact vulnerable ecosystems

and livelihoods. These findings highlight the trade-offs inherent in SRM, where global temperature stabilization comes at the cost of regional climate disparities, necessitating further investigation into circulation and precipitation feedbacks.

Global Precipitation

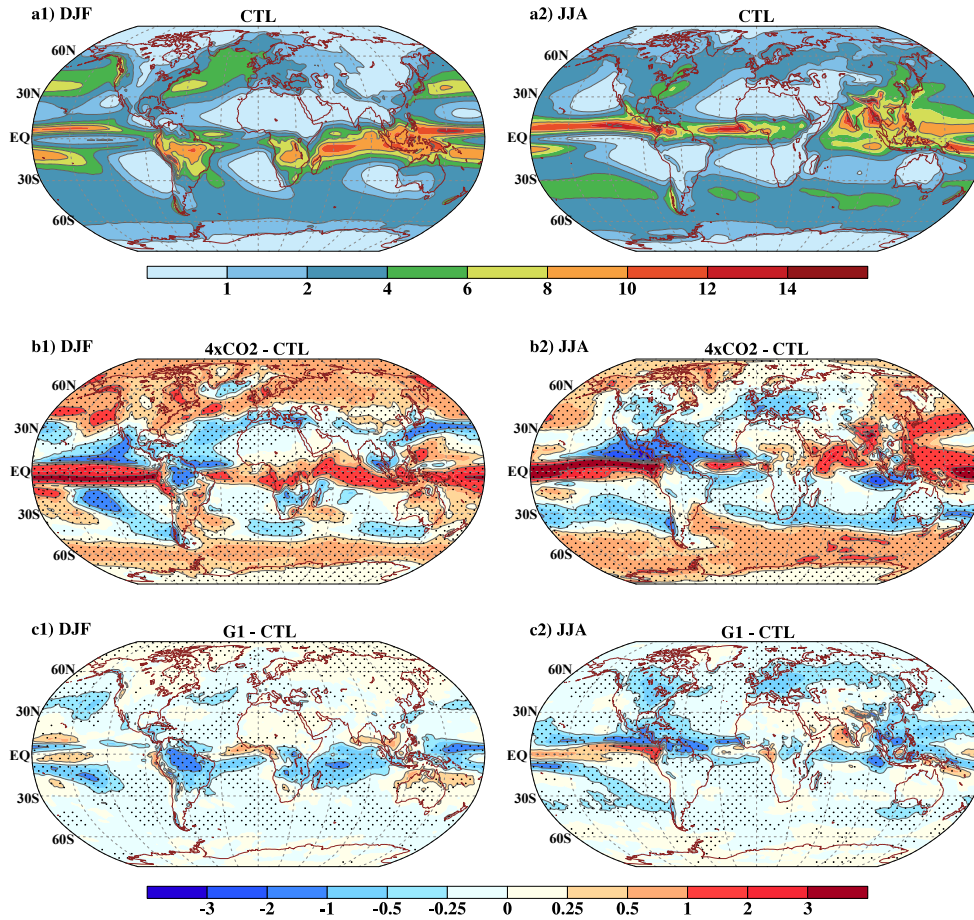


Figure 5: Spatial distribution of changes in precipitation in 4xCO₂ and G1 simulations w.r.t. control climate. Same as Fig. 4 but for precipitation from multi model mean of eight models of GeoMIP5.

The global precipitation response to 4xCO₂ relative to the CTL exhibits a spatially heterogeneous pattern, contrasting with the uniform warming observed in temperature fields (Figure 5, panels b1 and b2). Precipitation anomalies reveal both positive and negative changes, aligning with the "wet-get-wetter, dry-get-drier" paradigm driven by intensified hydrological cycling under elevated CO₂. Multi-model mean precipitation increases are most pronounced over equatorial regions and sub-polar lows, with enhancements exceeding 2 mm day⁻¹ in the deep tropics and 0.5–1 mm day⁻¹ in

sub-polar zones, reflecting strengthened meridional circulation and enhanced evaporation (Tilmes et al., 2013). Conversely, subtropical high-pressure zones experience reductions of 0.25–1.0 mm day⁻¹, representing 25–30% of climatological precipitation values, indicative of suppressed convection in these descending air regions (Figure 5, b1 and b2). Regionally, significant precipitation increases during boreal winter (DJF) are observed over eastern South America, northwestern Europe, northern Australia, and the rainforest regions of Central Africa, while decreases occur over the southern United States, Amazonia, and southern Europe (Figure 5, b1). During boreal summer (JJA), precipitation rises over South Asia, East Asia, and eastern Russia, but declines over Europe and the northwestern United States (Figure 5, b2). The increased moisture availability in a high-CO₂ world, due to enhanced evaporation over tropical oceans, underpins the amplified precipitation in low-latitude zones (Allen & Ingram, 2002).

Circulation-Driven Precipitation Changes in the 4xCO₂ Scenario

The precipitation anomalies in the 4xCO₂ scenario are closely linked to alterations in atmospheric circulation, as evidenced by wind field anomalies at 200 hPa and 850 hPa (Figures 3 and 4). During JJA, the Tibetan Anticyclone (TA) shifts northwestward, extending toward northwestern Europe, inducing a bifurcation of the upper-level (200 hPa) westerlies into northern and southern branches (Figure 6, b2). The southern branch merges into the TA, fostering an anomalous cyclonic circulation that weakens westerly strength, while the northern branch deflects toward polar regions (Simpson et al., 2019).

In contrast, during DJF, the absence of a prominent subtropical feature like the TA allows precipitation changes to follow westerly variations at both levels. Stronger westerlies north of Europe enhance cyclonic activity, increasing precipitation, while an anomalous anticyclonic circulation to the south induces a precipitation deficit (Figure 6, b1). A consistent reduction in precipitation along the subtropical eastern Pacific and Atlantic coasts of the United States is linked to the westward shift of the Azores/Bermuda High, associated with an anomalous anticyclonic circulation encircling the region (Figure 6, b2). This shift suppresses moisture influx, exacerbating drying in the northeastern United States during JJA (Figure 5, c2). The "wet-get-wetter" pattern is evident in high-precipitation zones such as the Intertropical Convergence Zone (ITCZ), South Pacific Convergence Zone (SPCZ), storm tracks, and the South Asian monsoon, with the notable

exception of Amazonia, which exhibits a reversed trend due to localized circulation feedbacks (Trisos et al., 2018)

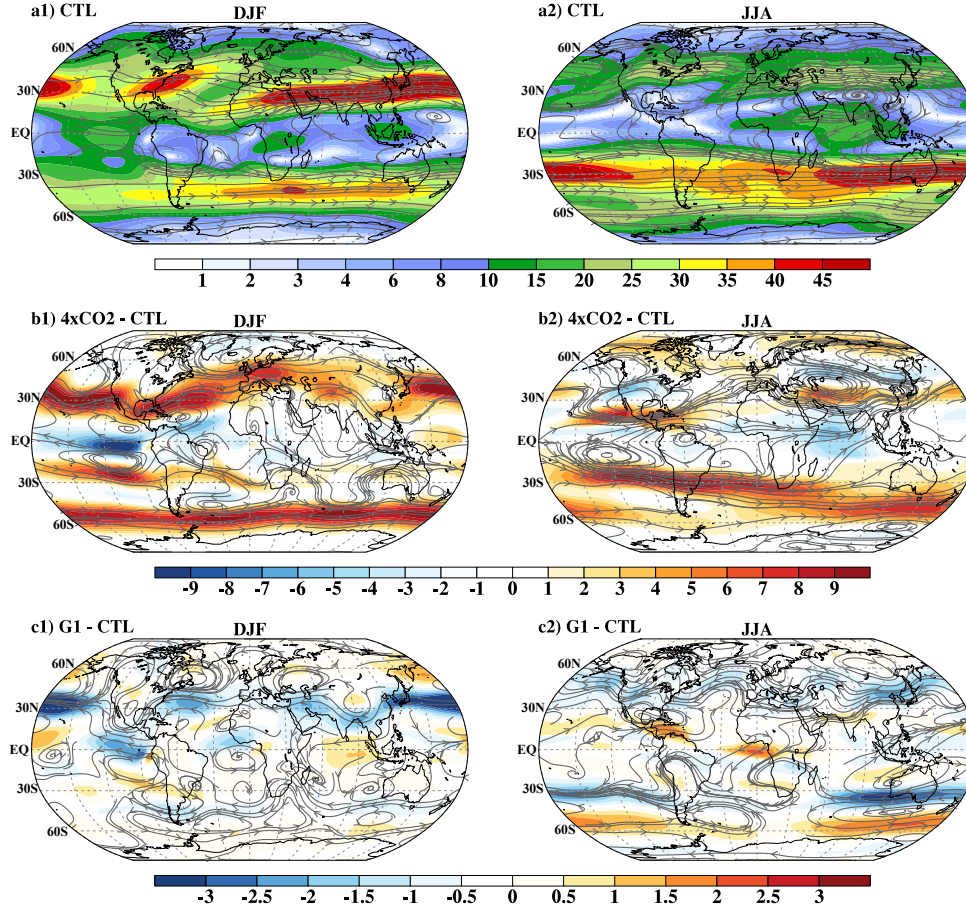


Figure 6: Spatial distribution of changes in winds at 200 hPa in 4CO₂ and G1 simulations w.r.t. CTL. (a1,a2) Winds at 200 hPa in CTL during (a1) winter and (b1) summer; the vectors and background shading represent the wind streamlines and wind speed respectively. (b1,b2) Differences in winds between 4CO₂ and CTL during (b1) winter and (b2) summer from multi model mean of eight models of GeoMIP5. (c1,c2) Same as (b1,b2) but for G1. The background shading in (b1,b2,c1,c2) represents the differences in wind speed between the experiment and the control climate.

The implementation of solar geoengineering in the G1 scenario significantly alters the precipitation regime, reflecting a weakened global hydrological cycle (Figure 5, panels c1 and c2). Multi-model mean precipitation reductions of 0.25–1 mm day⁻¹ are observed across most regions, consistent with suppressed evaporation and moisture transport under reduced solar insolation (Tilmes et al., 2013). During DJF, over-reduction of precipitation is notable along the ITCZ,

particularly over the Amazon rainforest in South America and the rainforest and savanna regions of Central Africa (Figure 5, c1). In JJA, significant decreases occur over eastern Canada, the northeastern United States, Mexico, Central America, northern South America, vast stretches of Europe and Russia, and Southeast Asia (Figure 5, c2). The northwestward displacement of the TA in the 4xCO₂ scenario persists partially in the G1 climate, with a residual anomalous anticyclonic circulation over central Asia (Figure 6, c2). This shift, combined with a consistent weakening of westerlies across the 30°N–60°N latitudinal belt at 200 hPa reflects changes in the meridional temperature gradient induced by SRM (Simpson et al., 2019). These circulation alterations contribute to the extensive precipitation deficits over the United States and Europe, likely stemming from suppressed mid-latitude storm tracks and reduced moisture convergence. The weakened westerlies and altered high-pressure systems underscore SRM's impact on dynamic climate features, amplifying regional disparities.

South Asian Monsoon response to SAI and RCP8.5 scenario

The spatial distribution of mean precipitation anomalies over the South Asian Monsoon (SAM) region under future climate scenarios is illustrated in Figure 7, comparing the Representative Concentration Pathway 8.5 (RCP8.5) and SAI scenarios relative to the CTL for the period 2070–2090 versus 2010–2030. The model, which interactively simulates aerosols in both the stratosphere and troposphere, captures the seasonal cycle of SAM rainfall reasonably well against observational datasets, though a mean dry bias is present, consistent with CMIP5/6 models (Taylor et al., 2012; Eyring et al., 2016).

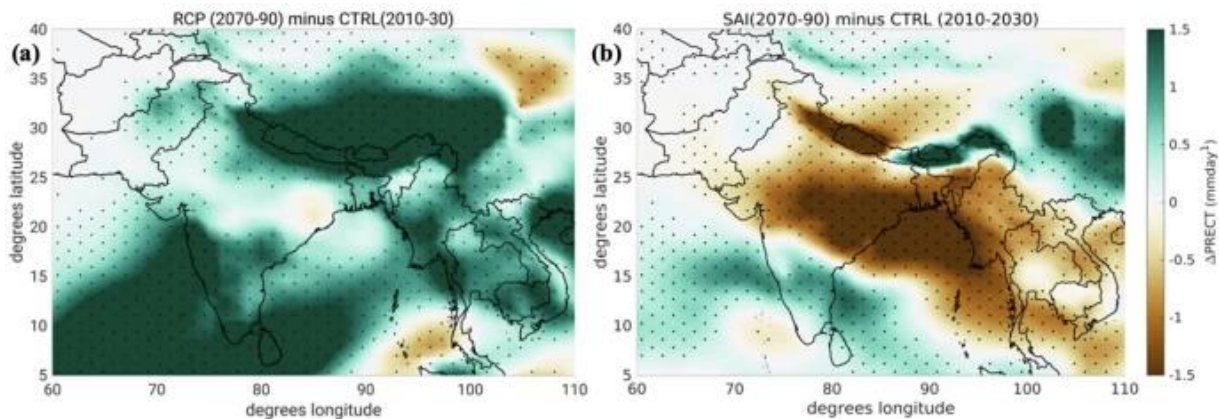


Figure 7: Difference in the JJA averaged precipitation between a RCP8.5 vs CTRL and b SAI vs CTRL in 2070–2090 vs 2010–2030. Stippling indicates the regions where the differences are significant for the difference of the mean (p-value < 0.05) using a two-sided t-test.

Under the RCP8.5 scenario, simulating extreme global warming, JJA (June–July–August) mean rainfall increases significantly across the Indian subcontinent, with notable enhancements over the Himalayas, northeast of the Bay of Bengal, and the west coast of India (Figure 7a). These increases, ranging from 5% over the core central Indian region (CI, as defined by Goswami et al., 2006) to 15% over the broader Indian Summer Monsoon Region (ISMR) compared to CTRL, are consistent with CMIP5/6 projections under SSP5-8.5 and RCP8.5 scenarios (Turner & Annamalai, 2012; Krishnan et al., 2020). This amplification is primarily attributed to the enhanced moisture-holding capacity of a warmer atmosphere, driven by increased evaporation over tropical oceans (Allen & Ingram, 2002).

In contrast, the SAI scenario reveals a marked decline in JJA rainfall across northern and central India, with reductions exceeding 1.5 mm day^{-1} in the CI region (Figure 7b). This decline extends eastward into the Bay of Bengal, with an area-averaged precipitation reduction greater than 20% over CI and approximately 11.6% over the ISMR compared to CTRL, indicative of drought-like conditions. However, precipitation increases are observed over southern India, extending northeast into parts of China, suggesting a regional redistribution of moisture (Figure 1b). The mechanisms driving these SAI-induced changes remain poorly understood, necessitating further investigation into circulation and thermodynamic feedbacks.

The precipitation anomalies under SAI are closely tied to changes in atmospheric circulation and vertical motions, but these are still left to be studied. Cloud dynamics play a pivotal role in modulating the energy and hydrological cycles under SAI. The model simulates a significant and widespread reduction in both low and high cloud cover over the Indian region, with high cloud cover decreasing by more than 5% in areas aligned with maximum rainfall reductions. This reduction mirrors the precipitation decline, indicating a feedback loop where diminished cloud formation suppresses convection and rainfall (Ramanathan et al., 2001). Low cloud anomalies exhibit a north-south dipole pattern similar to precipitation changes, with reductions over northern India and increases over southern regions, reinforcing the regional variability induced by SAI. The total cloud cover reduction further amplifies the drying effect, highlighting the interconnectedness of cloud, precipitation, and circulation responses in a geoengineered climate.

5. Discussions

The results demonstrate that solar geoengineering, as implemented in the G1 scenario, effectively mitigates the substantial warming induced by a quadrupling of atmospheric CO₂ concentrations, reducing global mean temperature anomalies to a residual range of 0.5–4°C compared to the 2.9–5.9°C increase in the 4xCO₂ scenario. This alleviation is consistent across regions, suggesting that no single area suffers disproportionately at the expense of others' benefits, a critical consideration for equitable climate intervention (Caldeira et al., 2013). However, the magnitude of residual warming varies spatially, with near-complete temperature mitigation in Asia and Africa, contrasted by higher residuals (up to 2°C) in Northern Hemisphere (NH) high latitudes, Europe, the United States, and Russia. This disparity reflects SRM's differential efficacy, driven by land-ocean heat contrasts and polar amplification feedbacks that persist despite insolation reduction (Serreze & Barry, 2011).

Precipitation responses further highlight regional inequities. The 4xCO₂ scenario exacerbates a "wet-get-wetter, dry-get-drier" pattern, intensifying hydrological disparities. Geoengineering mitigates this to some extent by weakening the global hydrological cycle, reducing precipitation by 0.25–1 mm day⁻¹ across most regions. However, over-drying in Europe, the United States, and Russia during boreal summer (JJA) contrasts with more stable conditions in South Asia, Africa, and Australia, where precipitation changes are less pronounced. This suggests that while SRM alleviates the extreme precipitation imbalances of a high-CO₂ world, it introduces new disparities, particularly in mid-latitude zones, where weakened westerlies and shifted circulation patterns suppress moisture availability (Simpson et al., 2019).

The marked decline in JJA rainfall over northern and central India, exceeding 1.5 mm day⁻¹ and reducing area-averaged precipitation by over 20% in the core central Indian region (CI) and 11.6% in the Indian Summer Monsoon Region (ISMR), contrasts sharply with the 5–15% precipitation increase under RCP8.5. This suggests that SAI, while stabilizing temperatures, exacerbates drought-like conditions in key agricultural zones, challenging the notion of uniform climate benefits (Caldeira et al., 2013).

The observed changes in meridional temperature gradients under the G1 scenario directly influence atmospheric circulation, notably the consistent weakening of westerlies across the 30°N–60°N latitudinal belt at both 200 hPa. This weakening, coupled with the northwestward displacement of the Tibetan Anticyclone (TA) and its residual anomalous anticyclonic motion over central Asia, disrupts traditional mid-latitude storm tracks and moisture transport. The bifurcation of upper-level westerlies and the formation of anomalous high-pressure systems over northwestern Europe explain the significant precipitation deficits in these regions during JJA. Similarly, the westward shift of the Azores/Bermuda High contributes to drying along the eastern United States, underscoring the role of dynamic circulation in amplifying regional disparities.

In South Asia, the partial relocation of the TA and overcooling over the Indo-Gangetic Plain during JJA suggest potential disruptions to the Indian Summer Monsoon, a critical system for regional water security (Bhowmick et al., 2021). The lack of statistical significance in precipitation changes over South Asia under G1 indicates model uncertainty, highlighting the need for higher-resolution studies to assess monsoon stability. These circulation-driven effects emphasize that SRM's global cooling benefit comes with trade-offs, particularly in regions dependent on specific climatic features.

The findings align with GeoMIP studies, which report a 4.5% global precipitation reduction under SRM due to suppressed evaporation (Tilmes et al., 2013), and with Vioni et al. (2023), who note circulation shifts like the ITCZ displacement as key drivers of regional disparities. The residual warming in NH high latitudes and the Arctic corroborates Schmidt et al. (2012), who attribute this to unmitigated CO₂ feedbacks, while the overcooling in tropical oceans reflects efficient radiative forcing reduction in low latitudes. The "wet-get-wetter" pattern in high-precipitation zones under 4xCO₂, except in Amazonia, is consistent with Trisos et al. (2018), who highlight localized circulation feedbacks.

Implications for Climate Strategy and Equity

The slow pace of global emission reduction efforts, constrained by geopolitical complexities and the infeasibility of rapid energy and transportation system transformations, underscores the urgency of exploring SRM as a complementary strategy (Rogelj et al., 2016). The geoengineered climate's closer resemblance to pre-industrial conditions, with mitigated temperature extremes and

alleviated precipitation disparities, positions SRM as a promising tool to avert catastrophic warming. However, the persistent over-drying in mid-latitude regions like Europe and the United States, contrasted with relative stability in South Asia and Africa, raises equity concerns, as developing nations may benefit more than industrialized ones (Rahman et al., 2018). Uncertainty remains a significant challenge, with potential risks including the "termination shock"—a rapid warming if SRM ceases— and unintended ecological impacts from circulation shifts (Matthews & Caldeira, 2007). The possibility that successful SRM deployment might reduce political will to curb emissions further complicates its adoption, potentially exacerbating long-term climate risks (Robock et al., 2008). To address these issues, global-scale engagement is essential, incorporating region-specific objectives and metrics to ensure equitable distribution of benefits and risks. This could involve tailored SRM deployment strategies, such as targeted aerosol injection, to minimize disparities, (Visioni et al., 2023).

Future Research Directions

There is a need for higher-resolution modeling to resolve uncertainties in monsoon regions like South Asia and to better capture aerosol microphysics and stratospheric feedbacks. Longitudinal analyses of SRM termination effects, incorporating carbon cycle dynamics, are critical to assess long-term stability. Additionally, integrating socio-economic data into climate models could enhance equity assessments, ensuring that SRM strategies prioritize vulnerable populations. These efforts will be vital to refining SRM's role in a comprehensive climate mitigation portfolio.

6. Conclusion

This term paper has systematically evaluated the regional climate disparities induced by solar geoengineering, with a particular focus on SRM scenarios as simulated through the GeoMIP. The findings affirm SRM's capacity to mitigate global warming, reducing temperature anomalies to a residual range of 0.5–4°C under the G1 experiment, compared to the 2.9–5.9°C increase projected under 4xCO₂ conditions. This cooling effect is most pronounced in Asia and Africa, though persistent warming in Northern Hemisphere high latitudes and the Arctic highlights limitations in counteracting polar amplification feedbacks.

However, the study reveals significant regional disparities in hydrological responses, challenging the notion of uniform climate benefits. The 4xCO₂ scenario intensifies a "wet-get-wetter, dry-get-drier" pattern, with precipitation increases in equatorial and sub-polar zones offset by reductions in subtropical highs. SRM, while weakening the global hydrological cycle, introduces new inequities, notably over-drying in mid-latitude regions like Europe and the United States, and a marked precipitation decline in northern India under SAI, exceeding 1.5 mm day⁻¹ and reducing monsoon rainfall by over 20% in the core central Indian region. These changes are driven by altered atmospheric circulation, including the northwestward shift of the Tibetan Anticyclone, weakened westerlies, and anomalous subsidence, which suppress convection and redistribute moisture regionally.

For South Asia, the Indian Summer Monsoon emerges as a critical vulnerability, with SAI-induced drought risks threatening water security and agriculture for over 1.4 billion people. The interplay of cloud feedbacks, vertical dynamics, and circulation shifts underscores the complexity of SRM's regional impacts, necessitating tailored strategies to mitigate adverse effects. While SRM offers a promising buffer against catastrophic warming amidst slow emission reduction progress, its potential to exacerbate inequities and the risk of termination shock demand cautious consideration.

The study's integration of GeoMIP5 and GeoMIP6 outputs provides a robust foundation for understanding these trade-offs, extending prior research by linking circulation changes to precipitation disparities. Future efforts should prioritize higher-resolution modeling to address uncertainties in monsoon dynamics, longitudinal analyses of termination risks, and socio-economic integration to ensure equitable outcomes. Ultimately, SRM's role in climate strategy hinges on global cooperation, region-specific governance, and a balanced approach that complements mitigation and adaptation, guiding humanity toward a resilient and just response to climate change.

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